

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE [PHASE-I & II]

TEST # 10

DATE : 03-11-2018

Test Type : Minor

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

Your Target is to secure Good Rank in AIIMS 2020

MINOR TEST # 10

Topics

DATE : 03 - 11 - 2018

PRE-MEDICAL NURTURE COURSE [PHASE-I & II] : [AIIMS]
PHYSICS : Work, Power & Energy, Circular Motion, Centre of Mass & Collision, Rotational Motion

CHEMISTRY : Periodic Table, Atomic Structure, Chemical Bonding, Thermodynamics, Chemical Equilibrium

BIOLOGY : Cell Structure, Cell Division, Anatomy of Flowering Plants [Botany]
 Digestive System, Respiratory System, Circulatory System,
 Excretory System [Zoology]

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- | | |
|--|--|
| <p>1. If a cyclist moving with a speed of 4.9 m/s on a level road can take a sharp circular turn of radius 4 m, then coefficient of friction between the cycle tyres and road is
 (1) 0.41 (2) 0.51 (3) 0.61 (4) 0.71</p> <p>2. The maximum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is
 (1) 60 (2) 30 (3) 15 (4) 25</p> <p>3. A motor cycle driver doubles its velocity when he is having a turn. The force exerted outwardly will be
 (1) Double (2) Half
 (3) 4 times (4) $\frac{1}{4}$ times</p> <p>4. A person with his hands in his pockets is skating on ice at the velocity of 10 m/s and describes a circle of radius 50 m. What is his inclination with vertical
 (1) $\tan^{-1}\left(\frac{1}{10}\right)$ (2) $\tan^{-1}\left(\frac{3}{5}\right)$
 (3) $\tan^{-1}(1)$ (4) $\tan^{-1}\left(\frac{1}{5}\right)$</p> <p>5. If the radius of curvature of the path of two particles of same masses are in the ratio $1 : 2$, then in order to have constant centripetal force, their velocity, should be in the ratio of
 (1) $1 : 4$ (2) $4 : 1$
 (3) $\sqrt{2} : 1$ (4) $1 : \sqrt{2}$</p> | <p>6. A car of mass 800 kg moves on a circular track of radius 40 m. If the coefficient of friction is 0.5, then maximum velocity with which the car can move is
 (1) 7 m/s (2) 14 m/s
 (3) 8 m/s (4) 12 m/s</p> <p>7. A mass is supported on a frictionless horizontal surface. It is attached to a string and rotates about a fixed centre at an angular velocity ω_0. If the length of the string and angular velocity are doubled, the tension in the string which was initially T_0 is now
 (1) T_0 (2) $T_0/2$
 (3) $4T_0$ (4) $8T_0$</p> <p>8. A bucket tied at the end of a 1.6 m long string is whirled in a vertical circle with constant speed. What should be the minimum speed so that the water from the bucket does not spill, when the bucket is at the highest position (Take)
 (1) 4 m/sec
 (2) 6.25 m/sec
 (3) 16 m/sec
 (4) None of the above</p> <p>9. A wheel is subjected to uniform angular acceleration about its axis. Initially its angular velocity is zero. In the first 2 sec, it rotates through an angle θ_1. In the next 2 sec, it rotates through an additional angle θ_2. The ratio of $\frac{\theta_2}{\theta_1}$ is :-
 (1) 1 (2) 2 (3) 3 (4) 5</p> |
|--|--|

SPACE FOR ROUGH WORK

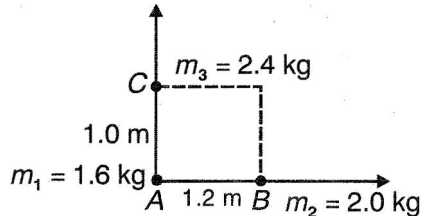
- | | |
|---|--|
| <p>10. A 1 kg stone at the end of 1 m long string is whirled in a vertical circle at constant speed of 4 m/sec. The tension in the string is 6 N, when the stone is at ($g = 10 \text{ m/sec}^2$)</p> <p>(1) Top of the circle
(2) Bottom of the circle
(3) Half way down
(4) None of the above</p> <p>11. A 2 kg stone at the end of a string 1 m long is whirled in a vertical circle at a constant speed. The speed of the stone is 4 m/sec. The tension in the string will be 52 N, when the stone is</p> <p>(1) At the top of the circle
(2) At the bottom of the circle
(3) Halfway down
(4) None of the above</p> <p>12. If the equation for the displacement of a particle moving on a circular path is given by $(\theta) = 2t^3 + 0.5$, where θ is in radians and t is in seconds, then the angular velocity of the particle after 2 sec from its start is</p> <p>(1) 8 rad/sec (2) 12 rad/sec
(3) 24 rad/sec (4) 36 rad/sec</p> <p>13. A uniform chain of length 2m is kept on a table such that a length of 60cm hangs freely from the edge of the table. The total mass of the chain is 4kg. What is the work done in pulling the entire chain on the table</p> <p>(1) 7.2 J (2) 3.6 J
(3) 120 J (4) 1200 J</p> <p>14. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle, the motion of the particle takes place in a plane. It follows that</p> | <p>(1) Its velocity is constant
(2) Its acceleration is constant
(3) Its kinetic energy is constant
(4) It moves in a straight line</p> <p>15. A force $\vec{F} = (5\hat{i} + 3\hat{j} + 2\hat{k})\text{N}$ is applied over a particle which displaces it from its origin to the point $\vec{r} = (2\hat{i} - \hat{j})\text{m}$. The work done on the particle in joules is</p> <p>(1) -7 (2) +7 (3) +10 (4) +13</p> <p>16. The potential energy of a certain spring when stretched through a distance 'S' is 10 joule. The amount of work (in joule) that must be done on this spring to stretch it through an additional distance 'S' will be</p> <p>(1) 30 (2) 40 (3) 10 (4) 20</p> <p>17. Two springs of spring constants 1500 N/m and 3000 N/m respectively are stretched with the same force. They will have potential energy in the ratio</p> <p>(1) 4 : 1 (2) 1 : 4
(3) 2 : 1 (4) 1 : 2</p> <p>18. A spring with spring constant k when stretched through 1 cm, the potential energy is U. If it is stretched by 4 cm. The potential energy will be</p> <p>(1) 4U (2) 8U (3) 16 U (4) 2U</p> <p>19. The spring extends by x on loading, then energy stored by the spring is :
(if T is the tension in spring and k is spring constant)</p> <p>(1) $\frac{T^2}{2k}$ (2) $\frac{T^2}{2k^2}$ (3) $\frac{2k}{T^2}$ (4) $\frac{2T^2}{k}$</p> |
|---|--|

SPACE FOR ROUGH WORK

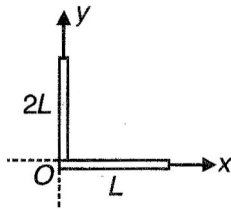
- | | |
|---|---|
| <p>20. The potential energy between two atoms in a molecule is given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$; where a and b are positive constants and x is the distance between the atoms. The atom is in stable equilibrium when</p> <p>(1) $x = \sqrt[6]{\frac{11a}{5b}}$ (2) $x = \sqrt[6]{\frac{a}{2b}}$</p> <p>(3) $x = 0$ (4) $x = \sqrt[6]{\frac{2a}{b}}$</p> <p>21. A bomb of mass 3.0 Kg explodes in air into two pieces of masses 2.0 kg and 1.0 kg. The smaller mass goes at a speed of 80 m/s. The total energy imparted to the two fragments is</p> <p>(1) 1.07 kJ (2) 2.14 kJ</p> <p>(3) 2.4 kJ (4) 4.8 kJ</p> <p>22. Two equal masses m_1 and m_2 moving along the same straight line with velocities + 3 m/s and – 5 m/s respectively collide elastically. Their velocities after the collision will be respectively</p> <p>(1) + 4 m/s for both</p> <p>(2) – 3 m/s and +5 m/s</p> <p>(3) – 4 m/s and + 4 m/s</p> <p>(4) – 5 m/s and + 3 m/s</p> <p>23. A body falling from a height of 10m rebounds from hard floor. If it loses 20% energy in the impact, then coefficient of restitution is</p> <p>(1) 0.89 (2) 0.56</p> <p>(3) 0.23 (4) 0.18</p> | <p>24. A tennis ball dropped from a height of 2 m rebounds only 1.5 m after hitting the ground. What fraction of its energy is lost in the impact</p> <p>(1) $\frac{1}{4}$ (2) $\frac{1}{2}$</p> <p>(3) $\frac{1}{3}$ (4) $\frac{1}{8}$</p> <p>25. A body of mass m moving with velocity v makes a head-on collision with another body of mass 2 m which is initially at rest. The loss of kinetic energy of the colliding body (mass m) is</p> <p>(1) $\frac{1}{2}$ of its initial kinetic energy</p> <p>(2) $\frac{1}{9}$ of its initial kinetic energy</p> <p>(3) $\frac{8}{9}$ of its initial kinetic energy</p> <p>(4) $\frac{1}{4}$ of its initial kinetic energy</p> <p>26. A particle of mass m moving with velocity v strikes a stationary particle of mass 2m and sticks to it. The speed of the system will be</p> <p>(1) $v/2$ (2) $2v$</p> <p>(3) $v/3$ (4) $3v$</p> |
|---|---|

SPACE FOR ROUGH WORK

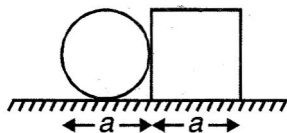
27. Three point masses m_1 , m_2 and m_3 are placed at the corners of a thin massless rectangular sheet ($1.2 \text{ m} \times 1.0 \text{ m}$) as shown. Centre of mass will be located at the point



- (1) $(0.8, 0.6) \text{ m}$ (2) $(0.6, 0.8) \text{ m}$
(3) $(0.4, 0.4) \text{ m}$ (4) $(0.5, 0.6) \text{ m}$
28. Figure shows a composite system of two uniform rods of lengths as indicated. Then the coordinates of the centre of mass of the system of rods are

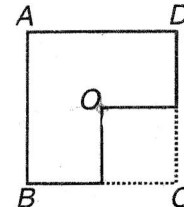


- (1) $\left(\frac{L}{2}, \frac{2L}{3}\right)$ (2) $\left(\frac{L}{4}, \frac{2L}{3}\right)$
(3) $\left(\frac{L}{6}, \frac{2L}{3}\right)$ (4) $\left(\frac{L}{6}, \frac{L}{3}\right)$
29. A circular plate of diameter 'a' is kept in contact with a square plate of side a as shown. The density of the material and the thickness are same everywhere. The centre of mass of composite system will be



- (1) Inside the circular plate
(2) Inside the square plate
(3) At the point of contact
(4) Outside the system

30. From a uniform square plate, one-fourth part is removed as shown. The centre of mass of remaining part will lie on



- (1) OC (2) OA (3) OB (4) OD
31. Two particles A and B initially at rest move towards each other under a mutual force of attraction. At the instant when velocity of A is v and that of B is $2v$, the velocity of centre of mass of the system
- (1) v (2) $2v$ (3) $3v$ (4) Zero
32. A ball of mass m is thrown upward and another ball of same mass is thrown downward so as to move freely under gravity. The acceleration of centre of mass is
- (1) g (2) $\frac{g}{2}$ (3) $2g$ (4) 0
33. A man of mass m starts moving on a plank of mass M with constant velocity v with respect to plank. If the plank lies on a smooth horizontal surface, then velocity of plank with respect to ground is

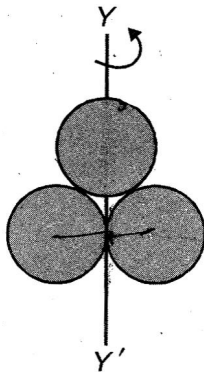
- (1) $\frac{Mv}{m+M}$ (2) $\frac{mv}{M}$
(3) $\frac{Mv}{m}$ (4) $\frac{mv}{m+M}$

SPACE FOR ROUGH WORK

34. A wheel starts from rest and attains an angular velocity of 20 radian/s after being uniformly accelerated for 10 s. The total angle in radian through which it has turned in 10 second is

(1) 20π (2) 40π
(3) 100 (4) 100π

35. Three solid spheres each of mass P and radius Q are arranged as shown in fig. The moment of inertia of the arrangement about YY' axis



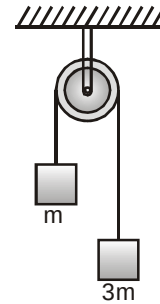
(1) $\frac{7}{5}PQ^2$ (2) $\frac{14}{5}PQ^2$
(3) $\frac{16}{5}PQ^2$ (4) $\frac{5}{14}PQ^2$

36. Three rods each of mass m and length L are joined to form an equilateral triangle as shown in the figure. What is the moment of inertia about an axis passing through the centre of mass of the system and perpendicular to the plane?



(1) $2mL^2$ (2) $\frac{mL^2}{2}$
(3) $\frac{mL^2}{3}$ (4) $\frac{mL^2}{6}$

37. If the system is released, then the acceleration of the centre of mass of the system is :-



(1) $\frac{g}{4}$ (2) $\frac{g}{2}$ (3) g (4) 2g

38. The moment of inertia of a body does not depend on :-

(1) the mass of the body
(2) the angular velocity of the body
(3) the axis of rotation of the body
(4) the distribution of the mass in the body

39. The moment of inertia of a circular ring of radius R and mass M about a tangent in its plane is :-

(1) MR^2 (2) $(1/2)MR^2$
(3) $(3/2)MR^2$ (4) $2MR^2$

40. The moment of inertia of a straight thin rod of mass M, length L about an axis perpendicular to its length and passing through its one end is :-

(1) $\frac{1}{12}ML^2$ (2) $\frac{1}{3}ML^2$
(3) $\frac{1}{2}ML^2$ (4) ML^2

SPACE FOR ROUGH WORK

41. The wavelengths of electron waves in two orbits is 3 : 5. The ratio of kinetic energy of electrons will be:
(1) 25 : 9 (2) 5 : 3
(3) 9 : 25 (4) 3 : 5
42. Which of the following electronic configurations, an atom has the lowest ionisation enthalpy?
(1) $1s^2 2s^2 2p^3$ (2) $1s^2 2s^2 2p^6 3s^1$
(3) $1s^2 2s^2 2p^6$ (4) $1s^2 2s^2 2p^5$
43. How many d-electrons in Cu^+ (At. No. = 29) can have the spin quantum number $\left(-\frac{1}{2}\right)$?
(1) 3 (2) 7 (3) 5 (4) 9
44. Which of the following sets of quantum numbers represents the highest energy of an atom?
(1) $n = 3, l = 0, m = 0, s = +\frac{1}{2}$
(2) $n = 3, l = 1, m = 1, s = +\frac{1}{2}$
(3) $n = 3, l = 2, m = 1, s = +\frac{1}{2}$
(4) $n = 4, l = 0, m = 0, s = +\frac{1}{2}$
45. According to Bohr theory, the angular momentum of electron in 5th orbit is:
(1) $25\frac{h}{\pi}$ (2) $1\frac{h}{\pi}$
(3) $10\frac{h}{\pi}$ (4) $2.5\frac{h}{\pi}$
46. The number of radial nodes of 3s- and 2p-orbitals are respectively :-
(1) 2, 0 (2) 0, 2
(3) 1, 2 (4) 2, 1
47. Which of the following statements in relation to the hydrogen atom is correct?
(1) 3s-orbital is lower in energy than 3p - orbital
(2) 3p-orbital is lower in energy than 3d-orbital
(3) 3s-and 3p-orbitals are of lower energy than 3d-orbital
(4) 3s, 3p and 3d-orbitals all have the same energy
48. 5 moles of an ideal gas at 27°C expands isothermally and reversibly from a volume of 6 L to 60 L. The work done in kJ is :
(1) -14.7 (2) -28.72
(3) +28.72 (4) -56.72
49. 1 litre-atm work is approximately equal to :-
(1) 101.3 J (2) 8.314 J
(3) 931 J (4) 19.2 J
50. A system absorbs 20 kJ heat and also does 10 kJ of work. The net internal energy of the system:
(1) increases by 10 kJ
(2) decreases by 10 kJ
(3) increases by 30 kJ
(4) decreases by 30 kJ

SPACE FOR ROUGH WORK

51. The standard molar heat of formation of ethane, CO_2 and $\text{H}_2\text{O}(l)$ are respectively -21.1 , -94.1 and -68.3 kcal. The standard molar heat of combustion of ethane will be :-

- (1) -372 kcal (2) -240 kcal
(3) 162 kcal (4) 183.5 kcal

52. ΔH_f° 298 K of methanol is given by the chemical equation :-

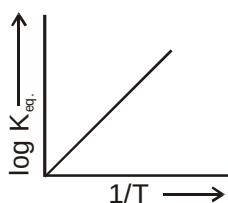
- (1) $\text{CH}_4(g) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{CH}_3\text{OH}(g)$
(2) $\text{C}(\text{graphite}) + \frac{1}{2}\text{O}_2(g) + 2\text{H}_2(g) \longrightarrow \text{CH}_3\text{OH}(l)$
(3) $\text{C}(\text{diamond}) + \frac{1}{2}\text{O}_2(g) + 2\text{H}_2(g) \longrightarrow \text{CH}_3\text{OH}(l)$
(4) $\text{CO}(g) + 2\text{H}_2(g) \longrightarrow \text{CH}_3\text{OH}(l)$

53. At 0°C , ice and water are in equilibrium and enthalpy change for the process

$\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(l)$ is 6 kJ mol^{-1} . Calculate the entropy change for the conversion of ice into liquid water.

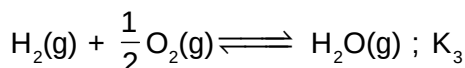
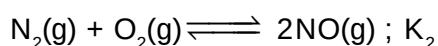
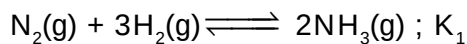
- (1) $0.02198 \text{ JK}^{-1} \text{ mol}^{-1}$
(2) $21.98 \text{ JK}^{-1} \text{ mol}^{-1}$
(3) $21.98 \text{ kJ K}^{-1} \text{ mol}^{-1}$
(4) $219.8 \text{ JK}^{-1} \text{ mol}^{-1}$

54. A schematic plot of $\log K_{\text{eq}}$ vs inverse of temperature for a reaction is shown in the figure. The reaction must be:

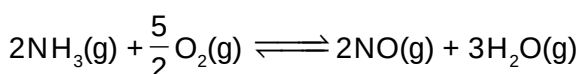


- (1) exothermic
(2) endothermic
(3) one with negligible enthalpy change
(4) highly spontaneous at ordinary temperature

55. Given :



The equilibrium constant for



will be :-

- (1) $K_1 K_2 K_3$ (2) $\frac{K_1 K_2}{K_3}$
(3) $\frac{K_1 K_3^2}{K_2}$ (4) $\frac{K_2 K_3^3}{K_1}$

56. For the reaction, $\text{N}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{NO}(g)$ the value of K_c at 800°C is 0.1 . When the equilibrium concentrations of both the reactants is 0.5 mol , what is the value of K_p at the same temperature?

- (1) 0.5 (2) 0.1
(3) 0.01 (4) 0.025

57. $\text{A}(g) + 3\text{B}(g) \rightleftharpoons 4\text{C}(g)$ initial concentration of A is equal to that of B. The equilibrium concentrations of A and C are equal. K_c of the reaction will be:

- (1) 0.08 (2) 0.8
(3) 8 (4) 80

SPACE FOR ROUGH WORK

58. For a hypothetical reaction :
 $4A(g) + 5B(g) \rightleftharpoons 4P(g) + 6Q(g)$ The equilibrium constant K_c has units:
 (1) mol L^{-1} (2) $\text{mol}^{-1} \text{L}$
 (3) $(\text{mol L}^{-1})^{-2}$ (4) unitless
59. Consider the following reversible reaction at equilibrium:
 $2H_2O(g) \rightleftharpoons 2H_2(g) + O_2(g); \Delta H = + 24.7 \text{ kJ}$
 which one of the following changes in conditions will lead to maximum decomposition of $H_2O(g)$?
 (1) Increasing both temperature and pressure
 (2) Decreasing temperature and increasing pressure
 (3) Increasing temperature and decreasing pressure
 (4) Increasing temperature at constant pressure
60. At constant pressure, the addition of argon to :
 $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ will :
 (1) shift the equilibrium in forward direction
 (2) shift the equilibrium in backward direction
 (3) not affect the equilibrium
 (4) stop the reaction
61. The vapour density of PCl_5 is 104.25 but when heated to 230°C , its vapour density is reduced to 62. The degree of dissociation of PCl_5 at this temperature will be :
 (1) 6.8% (2) 68% (3) 46% (4) 64%
62. The cause of diagonal relationship is
 (1) similar electronegativities
 (2) similar ionic or atomic radii
 (3) similar polarizing power of ions
 (4) all the three.
63. The atomic radii in case of inert gases is
 (1) ionic radii
 (2) covalent radii
 (3) van der Waals radii
 (4) none.
64. Which of the following combinations contains only isoelectronic ions?
 (1) N^{3-} , O^{2-} , Cl^- , Ne (2) F^- , Ar, S^{2-} , Cr.
 (3) P^{3-} , S^{2-} , Cl^- , Ar (4) N^{3-} , F^- , O^{2-} , Ar.
65. The ionic species having largest size is
 (1) $Li^+(g)$ (2) $Na^+(aq)$
 (3) $Rb^+(aq)$ (4) $Li^+(aq)$
66. The correct order of second ionization enthalpy of carbon, nitrogen, oxygen and fluorine is
 (1) $C > O > F > N$ (2) $O > N > F > C$
 (3) $O > F > N > C$ (4) $F > O > N > C$.
67. Which one of the following statements is incorrect?
 (1) Greater the nuclear charge, more negative is the electron gain enthalpy
 (2) Nitrogen has positive electron gain enthalpy
 (3) Electron gain enthalpy decreases from fluorine to iodine in the group
 (4) Chlorine has highest electron gain enthalpy with negative sign.

SPACE FOR ROUGH WORK

68. The ionization enthalpy and electron gain enthalpy of an element are 13.0 eV and 3.8 eV respectively. Its electronegativity on the Pauling scale is
 (1) 4.0 (2) 3.5 (3) 3.0 (4) 2.8.
69. Which one of the following molecules has the smallest bond angle ?
 (1) NH_3 (2) PH_3
 (3) H_2Se (4) H_2S
70. Carbon suboxide (C_3O_2) has recently been shown as a component of the atmosphere of Venus. Which of the following formulation represents the correct ground state Lewis structure for carbon suboxide ?
 (1) :O:C::C:C:O:
 (2) :O::C::C:C::O:
 (3) $\text{:}\ddot{\text{O}}::\text{C}::\text{C}::\text{C}::\ddot{\text{O}}:$
 (4) O:C:C:C:O:
71. Number of sigma bonds in P_4O_{10} is
 (1) 6 (2) 7 (3) 17 (4) 16
72. In compounds of the type ECl_3 , when $\text{E} = \text{B}, \text{P}, \text{As}$ or Bi , the angles Cl-E-Cl are
 (1) $\text{B} > \text{P} = \text{As} = \text{Bi}$ (2) $\text{B} > \text{P} > \text{As} > \text{Bi}$
 (3) $\text{B} < \text{P} = \text{As} = \text{Bi}$ (4) $\text{B} < \text{P} < \text{As} < \text{Bi}$
73. The dipole moment of LiH is found to be $20 \times 10^{-29} \text{ C m}$. If the interatomic distance in LiH is $1.6 \text{ \AA}$, then the percent ionic character of Li-H bond is nearly
 (1) 80% (2) 60% (3) 50% (4) 40%
74. Two hybrid orbitals have a bond angle of 120° . The percentage of s character in the hybrid orbital is nearly
 (1) 25% (2) 33% (3) 50% (4) 66%.
75. In which of the following molecule, S atom does not assume sp^3 hybridisation ?
 (1) SO_4^{2-} (2) SF_4 (3) SF_2 (4) S_8
76. Which of the following are isoelectronic and iso-structural ?
 $\text{NO}_3^-, \text{CO}_3^{2-}, \text{ClO}_3^-, \text{SO}_3$
 (1) $\text{NO}_3^-, \text{CO}_3^{2-}$ (2) $\text{SO}_3, \text{NO}_3^-$
 (3) $\text{ClO}_3^-, \text{CO}_3^{2-}$ (4) $\text{CO}_3^{2-}, \text{SO}_3$
77. Specify the coordination geometry around and hybridisation of N and B atoms in a 1 : 1 complex of BF_3 and NH_3
 (1) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3
 (2) N : pyramidal, sp^3 ; B : pyramidal, sp^3
 (3) N : pyramidal, sp^3 ; B : planar, sp^2
 (4) N : pyramidal, sp^3 ; B : tetrahedral, sp^3
78. The radius and charge of each of the six ions are shown below :
- | Ion | A^+ | B^+ | C^{2+} | X^- | Y^- | Z^{2-} |
|-----------|--------------|--------------|-----------------|--------------|--------------|-----------------|
| Radius/nm | 0.14 | 0.18 | 0.15 | 0.14 | 0.18 | 0.15 |
- The ionic solids AX , BY and CZ are of the same lattice type. What is the correct order of their lattice energies placing the one with the highest numerical value first ?
 (1) $\text{AX} > \text{BY} > \text{CZ}$ (2) $\text{AX} > \text{CZ} > \text{BY}$
 (3) $\text{BY} > \text{CZ} > \text{AX}$ (4) $\text{CZ} > \text{AX} > \text{BY}$
79. Shape and hybridisation of IF_5 respectively are
 (1) Trigonal bipyramidal, sp^3d
 (2) See saw, sp^3d
 (3) Square pyramidal, sp^3d^2
 (4) Pentagonal pyramidal, sp^3d^3
80. The percentage of p-character in the orbitals forming P-P bonds in P_4 is
 (1) 25 (2) 33 (3) 50 (4) 75

SPACE FOR ROUGH WORK

- | | |
|--|---|
| <p>81. A cell has 20 picogram of DNA at G_1 phase which undergone Meiosis. What will be the amount of DNA in the product of Meiosis I and Meiosis II?</p> <p>(1) 20 pg and 10 pg respectively
 (2) 10 pg and 5 pg respectively
 (3) 10 pg and 10 pg respectively
 (4) 40 pg and 20 pg respectively</p> <p>82. Some events of meiosis are given below.
 (A) Double equatorial plate formation
 (B) Bouquet stage formation
 (C) Chiasmata formation
 (D) Splitting of centromere
 Find the stages where these events takes place.</p> <p>(1) Metaphase I, Leptotene, Diplotene, Anaphase I
 (2) Metaphase I, Leptotene, Diplotene, Anaphase II
 (3) Metaphase II, Pachytene, Zygotene, Anaphase II
 (4) Metaphase I, Pachytene, Zygotene, Metaphase II</p> <p>83. Statement A : Poleward movent of dyads occurs during anaphase I
 Statement B : Centriole duplication and DNA replication occur in G_2-phase</p> <p>(1) Only (A) is correct
 (2) Only (B) is correct
 (3) Both (A) and (B) correct
 (4) Both (A) and (B) are incorrect</p> | <p>84. Disjunction is</p> <p>(1) Chromosome aberration involving translocation
 (2) Separation of homologous chromosome at anaphase
 (3) Deletion of genes
 (4) Exchange of genetic material</p> <p>85. Number of chromatid at metaphase is</p> <p>(1) Two each in mitosis and meiosis
 (2) Two in mitosis and four in meiosis
 (3) Two in mitosis and one in meiosis
 (4) More than one option is correct</p> <p>86. All of statements are correct w.r.t. Meiosis, except</p> <p>(1) Formation of four haploid cells at the end of process
 (2) Pairing of homologous chromosomes and recombination between them
 (3) Two sequential cycle of cell division but single cycle of nuclear division
 (4) Meiosis produces genetic variations in the offspring</p> <p>87. Incorrect about lenticels :-</p> <p>(1) Guard cells regulate opening and closing
 (2) Perform exchange of gases
 (3) Transpiration also occurs
 (4) Absent in young stem</p> |
|--|---|

SPACE FOR ROUGH WORK

88. The diploid cell of rice plants has 10 pg DNA in G_1 phase. What would be the DNA content in the cell at G_2 - phase of interphase during cell cycle?

 - 20 pg
 - 30 pg
 - 5pg
 - 40 pg

89. Which of the following common to stele and bark?

 - Cortex
 - Secondary phloem
 - Secondary xylem
 - Pith

90. Mention the DNA content of the cell at G_2 , Prophase I and Metaphase II of Meiosis.

 - 4C, 4C, 2C
 - 2C, 1C, 2C
 - 4C, 2C, 1C
 - 4C, 2C, 2C

91. Dicot root is different from monocot root in absence of

 - Radial arrangement of vascular bundles
 - Cambium in between xylem and phloem
 - Larger and well-developed pith
 - Unicellular root hairs

92. Differential activity of cambium is observed in :-

 - Stem in tropical region
 - Stem in temperate region
 - Root in tropical region
 - Root in temperate region

93. Read the following five statements (a-d)

 - Bark is non technical term
 - Girdling is not possible in monocot stem
 - Secondary phloem and periderm are included in bark
 - In dicot root vascular cambium primary in origin

How many statement are wrong?

 - Two
 - Three
 - Four
 - One

94. How many of the following meristems are completely secondary in origin?

 - Cork cambium of a dicot stem
 - Vascular cambium of a dicot stem
 - Cork cambium of a dicot root
 - Vascular cambium of a dicot root

Options :

(1) 4	(2) 3
(3) 2	(4) 1

SPACE FOR ROUGH WORK

- | | |
|--|---|
| <p>95. Interfascicular cambium and cork cambium are secondary meristems because :-</p> <ol style="list-style-type: none"> (1) They appear early in the life of a plant and contribute to the formation of the primary plant body. (2) They appear early in the life of a plant and contribute to the formation of the secondary plant body. (3) They appear later than primary meristem and are responsible for producing the primary tissues (4) They appear later than primary meristem and are responsible for producing the secondary tissues. | <p>98. Statement-1 : Most distinct annual rings are formed in temperate region plants.</p> <p>Statement-2 : In temperate region, the climatic condition are not uniform through the year.</p> <ol style="list-style-type: none"> (1) Statement-1 and 2 both are true (2) Statement-1 is false, Statement-2 is true (3) Statement-1 is true, Statement-2 is false (4) Statement-1 and 2 both are false |
| <p>96. Palisade parenchyma in leaf is/has</p> <ol style="list-style-type: none"> (1) Adaxial placed elongated cells (2) Extend to the lower epidermis (3) Numerous large spaces present between these cells (4) Oval/round and loosely arranged parenchyma cells | <p>99. What is absent in monocot stem?</p> <ol style="list-style-type: none"> (1) Vascular cambium (2) Collenchyma in hypodermis (3) Pericycle and endodermis (4) All of these |
| <p>97. Select an incorrect statement about anatomy of leaves.</p> <ol style="list-style-type: none"> (1) Monocot leaves are amphistomatic (2) Dicot leaves are generally hypostomatic (3) Xylem is on adaxial surface of leaves (4) Mesophyll is well-differentiated into monocot leaves | <p>100. Extrastelar secondary growth takes place by :-</p> <ol style="list-style-type: none"> (1) Vascular cambium (2) Phellogen (3) Phellem (4) Phelloderm |

SPACE FOR ROUGH WORK

101. Which of following is not a common symptom in both PEM-diseases, marasmus and Kwashiorkor ?

- (1) Wasting of muscles
- (2) Protein deficiency
- (3) Failure of growth and brain development
- (4) Extensive oedema and swelling of body parts

102. What is common among amylase, rennin and trypsin?

- (1) these are produced in stomach
- (2) these act at a pH lower than 7
- (3) these all are proteins
- (4) these all are proteolytic enzymes

103. Read the following statements regarding ruminant stomachs. Which of the following is incorrect?

- (1) Compound stomach has four chambers, namely rumen, reticulum, omasum and abomasum
- (2) Rumen is the largest while reticulum is the smallest chamber
- (3) Rumen and reticulum are sites of cellulose digestion, where bacteria and protozoans carry out fermentation of cellulose
- (4) Omasum is the true stomach as it secretes gastric juice and hydrochloric acid

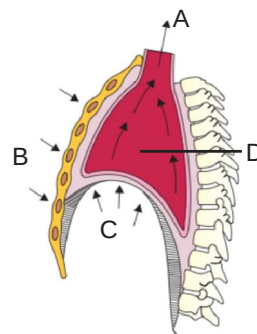
104. If pancreas is removed, the compound which remain undigested is :

- (1) proteins
- (2) carbohydrates
- (3) fats
- (4) all these

105. Read the following statements regarding hormones released in our gut. Which of the following is incorrect?

- (1) Gastrin is produced by G-cells of stomach and stimulates secretion of gastric juice
- (2) Secretin increase volume of pancreatic juice including water and electrolyse
- (3) Enterogastrone produced by small intestine slows down the secretion of gastric juice
- (4) Pancreozymin stimulates intestine to secrete the intestinal juice

106. Following diagram depicts the mechanism of breathing. In which of the following options, all the parts A, B, C and D are correctly labelled?



- (1) A-Air expelled from lungs
B-Ribs and sternum raised
C-Diaphragm relaxed
D-Volume of thorax decreased
- (2) A-Air expelled from lungs
B-Ribs and sternum raised
C-Diaphragm contracted
D-Volume of thorax decreased
- (3) A-Air expelled from lungs
B-Ribs and sternum lowered
C-Diaphragm relaxed and arched upwards
D-Volume of thorax decreased
- (4) A-Air entering into lungs
B-Ribs and sternum raised
C-Diaphragm contracted
D-Volume of thorax increased

107. Select the correct combination of the animal and its respiratory organ :-

Animal		Respiratory Structure	
(A)	Prawn	(i)	Dermal bronchiae
(B)	Earthworm	(ii)	Tracheae
(C)	Cockroach	(iii)	Moist skin
(D)	Scorpion	(iv)	Gills
(E)	Starfish	(v)	Book lungs

- (1) A-(v), B-(i), C-(ii), D-(iv), E-(iii)
- (2) A-(iv), B-(iii), C-(ii), D-(v), E-(i)
- (3) A-(iv), B-(ii), C-(iii), D-(i), E-(v)
- (4) A-(ii), B-(iii), C-(iv), D-(v), E-(i)

108. A person suffers from punctures in his chest cavity in an accident, without any damage to the lungs. Its effect could be :-

- (1) No change in breathing rate
- (2) Reduced breathing rate
- (3) First increases then reduced breathing rate
- (4) Cessation of breathing

109. Which of the following is incorrect?

- (1) P_{O_2} and P_{CO_2} of alveolar air is 104 mmHg and 40 mmHg respectively
- (2) P_{O_2} and P_{CO_2} of deoxygenated blood is 40 mmHg and 45 mmHg respectively
- (3) P_{O_2} and P_{CO_2} of exhaled air is 116 mmHg and 32 mmHg respectively
- (4) P_{O_2} and P_{CO_2} of inhaled air is 159 and 40 mmHg respectively

110. During inspiration, air moves into lungs because :-

- (1) Pressure within the lungs and atmospheric air is same
- (2) Pressure within lungs is more than atmospheric pressure
- (3) Pressure within lungs is less than the atmospheric pressure
- (4) Passive flow of air takes air inside

111. Which of the following statements is the most correct regarding excitation of heart muscles?

- (1) Only the nodal musculature has the ability to generate action potential
- (2) Only the SA node in heart has the ability to generate action potential
- (3) All the heart muscles have ability to auto excitation
- (4) Only the purkinje fibre is auto excitable centre of the heart

112. Which of the following factors is not always reported in RBC?

- (A) Antigen
- (B) Rh factor
- (C) Nucleus
- (D) Haemoglobin
- (E) Biconcave shape
- (1) (A), (B), (C) and (D)
- (2) (B), (C), (D) and (E)
- (3) (C), (D) and (E)
- (4) (A), (B), (C) and (E)

113. Which of the following is a matching pair?

- (1) Initiation of heart beat – Purkinje fibres
- (2) Pulsation of radial artery – Valves in blood vessels
- (3) Dup – Sudden opening of semilunar valves at the beginning of ventricular systole
- (4) Lubb – Sharp closure of AV valves at the beginning of ventricular systole

114. Which of the following statements is not correct about all vertebrate's heart?

- (1) Presence of muscular chambered heart
- (2) Presence of SA node
- (3) Presence of myogenic heart
- (4) It always pumps together oxygenated and deoxygenated blood to the different body parts.

115. Which of the following is correct?
- (1) Nearly 70% of blood is received by ventricles during joint diastole
 - (2) Auricular systole and ventricular systole never overlap
 - (3) The duration of joint diastole is 0.4 seconds.
 - (4) All are correct
116. Total thrombocytic count in human blood is :
- (1) 15,000 - 25,000
 - (2) 1,50,00 - 3,50,00
 - (3) 3,00,00 - 5,00,00
 - (4) 10,000 - 20,000
117. Conversion of fibrinogen in fibrin filaments is very important for blood clotting, which occurs by catalysis of :
- (1) Prothrombinase
 - (2) Thrombokinas
 - (3) Thrombin
 - (4) Prothrombin

118. Which of following is incorrect with respect to human heart ?
- (1) It is tetrachambered and myogenic
 - (2) Surrounded by pericardium
 - (3) Located in mediastinum
 - (4) Endodermal in origin
119. Select a correct statement about working of heart :
- (1) It beats 72 times per minute
 - (2) Pumps 70 ml blood in each aorta by single systole
 - (3) Produces lubb and dubb sound in each cardiac cycle
 - (4) All of these.
120. Myocardial ischemia is insufficient blood and O_2 supply to heart wall which is initially characterised by:
- (1) Cardiac arrest
 - (2) Angina pectoris
 - (3) Heart failure
 - (4) Hypertension

Direction for Assertion & Reason questions. These questions consist of two statements each, printed as assertion and reason. while answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

121. *Assertion* : Position of the centre of mass can lie outside the body also.
- Reason* : Centre of mass doesn't shift its position in the absence of an external force.

122. *Assertion* : When a rigid body rotates about any axis, then all the particles of it move in circles of different radii but with same angular velocity.
- Reason* : In rigid body relative position of particles are fixed.
123. *Assertion* : For the body to be in rotational equilibrium the net torque acting on the body about any point is zero.
- Reason* : For net torque to be zero, net force should also be zero.
124. *Assertion* : A rigid body can't be in a pure rolling on a rough inclined plane without giving any external force.
- Reason* : Since there is no torque providing force acting on the body in the above case, the body can't come in a rolling condition.
125. *Assertion* : When a ring moves in pure rolling condition on ground, it has 50% translational and 50% rotational energy.

Reason :

$$\frac{KE_{trans}}{KE_{rot}} = \frac{\frac{1}{2}MV^2}{\frac{1}{2}I\omega^2} = \frac{\frac{1}{2}MV^2}{\frac{1}{2}(MR^2)\frac{V^2}{R^2}} = 1:1$$

- | | |
|--|--|
| <p>126. <i>Assertion</i> : If both the speed of a body and radius of its circular path are doubled, then centripetal force also gets doubled.
<i>Reason</i> : Centripetal force is directly proportional to both speed of a body and radius of circular path.</p> <p>127. <i>Assertion</i> : In circular motion, work done by centripetal force is zero.
<i>Reason</i> : In circular motion centripetal force is perpendicular to the displacement.</p> <p>128. <i>Assertion</i> : During a turn, the value of centripetal force should be less than the limiting frictional force.
<i>Reason</i> : The centripetal force is provided by the frictional force between the tyres and the road.</p> <p>129. <i>Assertion</i> : A safe turn by a cyclist should neither be fast nor sharp.
<i>Reason</i> : The bending angle from the vertical would decrease with increase in velocity.</p> <p>130. <i>Assertion</i> : Cream gets separated out of milk when it is churned, it is due to gravitational force.
<i>Reason</i> : In circular motion gravitational force is equal to centripetal force.</p> <p>131. <i>Assertion</i> : A coin is placed on phonogram turn table. The motor is started, coin moves along the moving table.
<i>Reason</i> : The rotating table is providing necessary centripetal force to the coin.</p> <p>132. <i>Assertion</i> : A person working on a horizontal road with a load on his head does no work.
<i>Reason</i> : No work is said to be done, if directions of force and displacement of load are perpendicular to each other.</p> <p>133. <i>Assertion</i> : The work done during a round trip is always zero.
<i>Reason</i> : No force is required to move a body in its round trip.</p> | <p>134. <i>Assertion</i> : When a gas is allowed to expand, work done by gas is positive.
<i>Reason</i> : Force due to gaseous pressure and displacement (of piston) are in the same direction.</p> <p>135. <i>Assertion</i> : A light body and heavy body have same momentum. Then they also have same kinetic energy.
<i>Reason</i> : Kinetic energy does not depend on mass of the body.</p> <p>136. <i>Assertion</i> : A spring has potential energy, both when it is compressed or stretched.
<i>Reason</i> : In compressing or stretching, work is done on the spring against the restoring force.</p> <p>137. <i>Assertion</i> : Comets move around the sun in elliptical orbits. The gravitational force on the comet due to sun is not normal to the comet's velocity but the work done by the gravitational force over every complete orbit of the comet is zero.
<i>Reason</i> : Gravitational force is a non conservative force.</p> <p>138. <i>Assertion</i> : The power of a pump which raises 100 kg of water in 10sec to a height of 100 m is 10 KW.
<i>Reason</i> : The practical unit of power is horse power.</p> <p>139. <i>Assertion</i> : According to law of conservation of mechanical energy change in potential energy is equal and opposite to the change in kinetic energy.
<i>Reason</i> : Mechanical energy is not a conserved quantity.</p> <p>140. <i>Assertion</i> : Mass and energy are not conserved separately, but are conserved as a single entity called mass-energy.
<i>Reason</i> : Mass and energy conservation can be obtained by Einstein equation for energy.</p> |
|--|--|

141. *Assertion* : The electronic configuration of nitrogen atom is represented as



and not as



Reason : The electronic configuration of the ground state of an atom is the one which has the greatest multiplicity.

142. *Assertion* : Bohr's orbits are called stationary orbits.

Reason : Electrons remain stationary in these orbits for some time.

143. *Assertion* : The two lobes of a p-orbital have opposite sign.

Reason : The two lobes of a p-orbital lie on the opposite sides of the nucleus.

144. *Assertion* : Schrodinger wave equation helps as to find the probability of the electron in any orbital at different points around the nucleus.

Reason : All the four quantum numbers are obtained by solving Schrodinger wave equation.

145. *Assertion* : The free gaseous Cr atom has six unpaired electrons.

Reason : Half-filled s-orbital has greater stability.

146. *Assertion* : The thermodynamic function which determines the spontaneity of a process is the free energy. For a process to be spontaneous, the change in free energy must be negative.

Reason : The change in free energy is related to the change in enthalpy and change in entropy. The change in entropy for a process must always be positive if it is spontaneous.

147. *Assertion* : Heat of neutralization of HF is more than that of HCl by NaOH.

Reason : HF is stronger acid than HCl.

148. *Assertion* : The entropy of the universe is continuously increasing.

Reason : All naturally occurring processes are accompanied by increase in entropy.

149. *Assertion* : Internal energy change in a cyclic process is always + ve (i.e., > 0).

Reason : Internal energy is a path function.

150. *Assertion* : The active mass of a pure solid or a pure liquid is taken as 1.

Reason : The active mass of a pure solid or a pure liquid depends upon the density and molecular mass of the substance both of which are constant for the pure substance.

151. *Assertion* : If an inert gas is added to a dissociation equilibrium keeping pressure constant, the dissociation increases.

Reason : Inert gas exerts its own pressure in the vessel.

152. *Assertion* : K_p can be equal to or less than or greater than K_c .

Reason : $K_p = K_c (RT)^{\Delta n_g}$. Thus, K_p and K_c depend upon temperature.

153. *Assertion* : If reaction quotient (Q_c) is less than the equilibrium constant (K_c), the equilibrium tends to shift in the direction of products.

Reason : The expression for equilibrium constant is different than the expression for reaction quotient.

154. *Assertion* : Na^+ and Al^{3+} are isoelectronic but the magnitude of the ionic radius of Al^{3+} is less than that of Na^+

Reason : The magnitude of the effective nuclear charge on the outer shell electrons in Al^{3+} is greater than that in Na^+ .

155. *Assertion* : The single bond dissociation energy of F—F is less than that of O—O.

Reason : Each F-atom in F—F has three lone pairs of electrons while each O atom in O—O has two lone pairs.

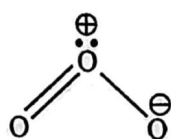
156. *Assertion* : All the P–F bonds in PF_5 do not have equal bond lengths.

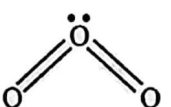
Reason : The hybridisation of P-atom involves not only s and p-orbitals but also d-orbitals.

157. *Assertion* : PCl_5 is not very stable and easily dissociates to form PCl_3 and Cl_2 .

Reason : The central atom (P) in PCl_5 has 10 electrons around it instead of 8, i.e., it violates octet rule.

158. *Assertion* : The electronic structure of O_3 is



Reason :  structure is not allowed because octet around O cannot be expanded.

159. *Assertion* : Noble gases have highest ionization enthalpies in their respective periods.

Reason : Noble gases have stable closed shell electronic configuration.

160. *Assertion* : The electron gain enthalpy of N is +ve while that of P is –ve.

Reason : This is due to the smaller atomic size of N in which there is a considerable electron-electron repulsion and hence the additional electron is not accepted easily.

161. *Assertion* : Bivalent to monovalent stage of homologous chromosome is concerned with anaphase I.

Reason : All chromosomes form a single equatorial plate in metaphase I.

162. *Assertion* : Meiosis does not occur in haploid cells.

Reason : All plants form gametes by meiosis.

163. *Assertion* : Mitotic division is an equational division.

Reason : Daughter cells are identical to each other as well as to parent cell.

164. *Assertion* : Cytokinesis is the division of cytoplasm.

Reason : In plant cells, cell wall formation starts in the centre of cell and grows outward to meet the existing lateral walls.

165. *Assertion* : Meiosis involves pairing of homologous chromosomes and recombination between them.

Reason : It produces genetic variability in the population of organisms from one generation to the next.

166. *Assertion* : Some cells left behind from shoot apical meristem during formation of leaves and elongation of stem which constitute axillary bud.

Reason : Such buds are present in the axile of leaves and capable of forming a branch or a flower.

167. *Assertion* : Simple permanent tissue form the major component within plant organs.

Reason : Parenchyma generally provide mechanical support to the growing part of plant.

168. *Assertion* : Lateral meristems are cylindrical meristem.

Reason : This meristems occur in mature region of root and shoots of many plants.

169. *Assertion* : Intercalary meristem is a portion of secondary meristem.

Reason : They are not present from beginning but develop at later stage through dedifferentiation of permanent tissue.

170. *Assertion* : Vascular bundles in dicot stem are conjoint, collateral and open.

Reason : In these bundles, xylem and phloem are present on same radius, phloem towards outer side and a strip of cambium separates the two.

171. <i>Assertion</i> : Enterokinase is secreted by the wall of duodenum. <i>Reason</i> : Enterokinase breaks down proteins into peptone and large peptidase. 172. <i>Assertion</i> : Presence of peptides and proteins in stomach stimulates the release of gastrin hormone by G-cells. <i>Reason</i> : Gastrin inhibits release of acidic gastric juice in stomach. 173. <i>Assertion</i> : 30% of starch can be digested in mouth in to maltose and isomaltose if food is properly chewed. <i>Reason</i> : Saliva contains β-amylase that can breakdown starch into disaccharidase. 174. <i>Assertion</i> : Trachea and bronchi do not collapse during expiration when pressure in them decrease. <i>Reason</i> : Linning of surfactant in trachea and bronchi reduces surface tension and prevents their collapse. 175. <i>Assertion</i> : Respiration involves the transport of gases by the blood. <i>Reason</i> : Gaseous exchange takes place across the respiratory membrane by diffusion.	176. <i>Assertion</i> : Inability of heart to pump sufficient blood is called heart failure. <i>Reason</i> : Patients suffer with breathlessness and dyspnea because heart failure causes congestion of lungs. 177. <i>Assertion</i> : Lymphocytes are phagocytic WBCs. <i>Reason</i> : Lymphocytes engulf pathogens and digest them with lysosomal enzymes. 178. <i>Assertion</i> : Blockage of coronary arteries by deposition of fats, cholesterol and fibres causes cerebral stroke. <i>Reason</i> : Raised levels of HDL increase the risk of atherosclerosis. 179. <i>Assertion</i> : Vital capacity is higher in athletes than non athletes. <i>Reason</i> : Vital capacity is about 3.5 - 4.5 ltrs in a normal adult person. 180. <i>Assertion</i> : During inspiration the diaphragm and external intercostal muscle contract simultaneously. <i>Reason</i> : Inspiration occurs due to muscular relaxation.
---	--



SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

ALLEN RESULT : NEET (UG) 2018

ALLEN MAINTAINS THE LEADERSHIP

With 55 Students in Top 100 AIR

#ItHappensOnlyAtALLEN

AIR - 2



ROHAN PUROHIT
Distance

AIR - 5



PRINCE CHOUDHARY
Classroom

AIR - 6



VARUN MUDDIDI
Distance

AIR - 7



AGARWAL KRISHNA
Distance

AIR - 9



MADHAVAN GUPTA
Distance

AIR - 10



RAMNEEK K. MAHAL
Distance

✓ **6 in TOP 10 All India Ranks**

✓ **11 in TOP 20 All India Ranks**

✓ **31 in TOP 50 All India Ranks**

✓ **55 in TOP 100 All India Ranks**

AIR - 11	AIR - 12	AIR - 13	AIR - 15	AIR - 20	AIR - 21	AIR - 22	AIR - 23	AIR - 24	AIR - 25	AIR - 26	AIR - 28	AIR - 29	AIR - 31	AIR - 34
ADITYA GUPTA Distance	KEERTHANA KASI Distance	RITVIK K. SAHOO Distance	SAHIL SHAH Distance	AMULYA GUPTA Classroom	GUNJAN A. GATTANI Distance	SANGEET RATH Classroom	ADITYA V. AGARWAL Distance	SHLOK V. PATEL Classroom	SIDDHARTH RAVI Distance	MOHD. ANAS Y. MEMON Classroom	SHIKHAR BANSAL Distance	SHABISTA KHAN Distance	MEHAK Classroom	AMITABH P. CHAUDHARY Classroom
AIR - 35	AIR - 37	AIR - 38	AIR - 41	AIR - 42	AIR - 43	AIR - 47	AIR - 48	AIR - 49	AIR - 50	AIR - 51	AIR - 52	AIR - 55	AIR - 58	AIR - 60
AVNIKA JAIN Distance	LOKESH MAHALECHA Distance	VIRENDRA CHOUDHARY Distance	ARJUN SARAWAT Distance	NISHTHA AGRAWAL Classroom	SANKALPA KESHARI Distance	PARUL WERRY Classroom	SAMAGRA SINGH Classroom	SADAR MISHRA Classroom	AARUSHI KOLLANA VISHWESH M. Distance	DHARA DIYA PRATHAM SINGH Distance	RAGHAVEN KHANDAL Distance	AYUSHI MITTAL Classroom	AYUSHI BANSAL Classroom	
AIR - 62	AIR - 65	AIR - 66	AIR - 69	AIR - 70	AIR - 73	AIR - 74	AIR - 75	AIR - 77	AIR - 78	AIR - 79	AIR - 81	AIR - 83	AIR - 84	AIR - 90
RIYA SINGH Classroom	PRATEEK K. MISHRA Classroom	VAIDHAV MITTAL Classroom	VAIDHAV GARG Distance	AYAN AGARWAL Distance	SANJEEV REDDY Distance	SAVI GADEGONE Distance	PRANJAL PRASAD Distance	PRAJWAL ANJIA Distance	PARYA N. LADANI Distance	PULKIT GUPTA Classroom	V.K.T.V. BELLAMKONDA Distance	PRABHJOT SINGH Classroom	RUTUOHWAJ SAVALIYA Classroom	ESHWAIK ADGARWAL Distance
AIR - 91	AIR - 95												AIR - 96	AIR - 100
SHUBHAM MARODIA Distance	KSHITIJ AGARWAL Classroom												JITENDRA YADAV Classroom	VIDUSHI VARSHNEY Classroom

81230*

Classroom
50267
Distance
30963

Highest Number of Students Qualified above Cut Off Marks among All Institutes of INDIA

Heartiest
Congratulations!



www.allen.ac.in

Authenticity of Result:
Power of **ALLEN**